

Question number	Answer	Additional guidance	Mark
4(a)	An answer that makes reference to the following points: • {protein / found in organisms} (1) • that {decreases activation energy / speeds up (the rate of) reaction} (without being used up) (1)	ACCEPT in our {bodies / cells / cytoplasm} / in a named organism ACCEPT description of activation energy	(2)

Question number	Answer	Mark
4(b)	The only correct answer is B (Q) <i>A is incorrect because Q is the activation energy</i> <i>C is incorrect because Q is the activation energy</i> <i>D is incorrect because Q is the activation energy</i>	(1)

Question number	Answer	Additional guidance	Mark
4(c)	An explanation that includes the following points: - substrate {binds / fits / forms enzyme-substrate complex} to the <u>active site</u> of the enzyme (1) - due to the <u>shape</u> of the {active site / enzyme / substrate} (1) - which is due to the {R groups / amino acids} (forming the active site) (1) - AB <u>and</u> AC {are a different shape to A and B / do not fit into E₁} (1)	ACCEPT are not complementary to (E ₁) enzyme	(3)

Question number	Answer	Additional guidance	Mark
4(d)	A description that includes the following points: - pH 4 is the {most suitable / best} pH (1) 30°C is the {most suitable / best} temperature (1) - pH 2 <u>and</u> 60°C (both) denature the enzyme (1)		(2)